

GROUNDEDHUMAN: IMPLICIT INTERACTION GROUNDING FOR DYNAMIC HUMAN-SCENE RECONSTRUCTION

Anonymous authors

Paper under double-blind review

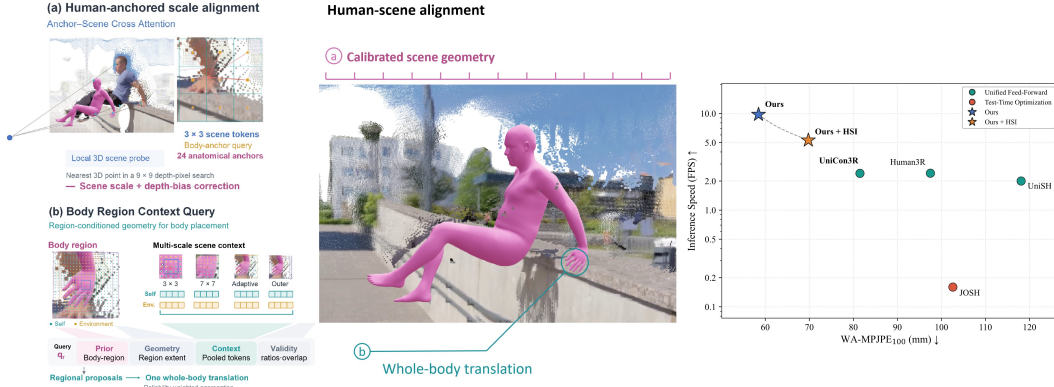


Figure 1: Overview of GroundedHuman. Left: body-anchored scene queries calibrate scene scale and depth bias, while body-region context queries aggregate multi-scale human-scene evidence to refine whole-body translation. Center: calibrated scene geometry and interaction-aware human placement. Right: world-space joint accuracy versus inference speed.

ABSTRACT

Reconstructing humans and their surroundings from monocular video requires a common metric scale and coherent human placement. These quantities are coupled: a local human-scene discrepancy may arise from either an incorrectly scaled scene or a misplaced body. We introduce GroundedHuman, which infers metric calibration and human placement from learned implicit interaction representations. Human structure determines how to query scene evidence; attention encodes their associations, and geometric readouts convert these relations into corrections. **Body-Anchored Metric Calibration** initializes scale from human surfaces and uses body-anchored scene queries to encode correspondence for residual calibration of scene depth and camera translation. Within this metric reference, **Dynamic Interaction Grounding** encodes regional human-environment relations through hierarchical context attention and reads out reliable whole-body translation updates. Each update changes the body’s relation to the scene, prompting the interaction representation to be recomputed while preserving pose and shape. This formulation grounds interaction without explicit contact prediction and reconstructs humans, scenes, and camera motion in a shared metric world. Experiments evaluate world-space human motion, scene depth, camera trajectories, and human-scene spatial consistency.

1 INTRODUCTION

Dynamic human-scene reconstruction from monocular video aims to recover how people move and interact within a shared physical environment. For applications such as augmented reality, embodied perception, and digital human animation, reconstructing the human and scene independently is insufficient: their scale, position, and motion must also be consistent in a common world coordinate

system. This requires jointly recovering human motion, scene geometry, and camera trajectories with physically meaningful spatial relationships (Chen et al., 2026; Li et al., 2026; Liu et al., 2026).

Two coupled ambiguities make this problem challenging. Modern scene reconstruction models recover accurate relative geometry and camera motion, but their predictions may lack a reliable metric scale, whereas parametric human estimators provide body geometry in metric units through learned human priors (Wang et al., 2025a; 2026a; Sáráandi & Pons-Moll, 2024). A mismatch between these references directly affects scene dimensions, camera translation, and world-space human motion. Yet metric consistency alone is not sufficient. Small errors in global human placement can still produce floating or penetration even when the human and scene share the same scale, while occlusion and incomplete geometry make the relevant interaction evidence difficult to identify.

Recent work has increasingly coupled human and scene reconstruction. Human3R jointly predicts humans, scenes, and cameras, while UniSH combines human and scene features for metric reconstruction and human placement (Chen et al., 2026; Li et al., 2026). UniCon3R introduces contact-aware feedback to correct human reconstruction, and GRAFT iteratively refines human parameters using nearby scene geometry (Sur et al., 2026; Pradyumna YM et al., 2026). These methods demonstrate the importance of scene-aware human reasoning, but metric ambiguity and interaction ambiguity remain closely entangled. Before local geometric discrepancies can reliably indicate whether a body is misplaced, the human and scene must first be expressed in a comparable metric space. Moreover, interaction constraints need not always be represented by explicit contact predictions; they can also be inferred from the spatial context surrounding the body.

Based on this observation, we introduce **GroundedHuman**, a framework that infers metric calibration and human placement from *implicit interaction representations*. Our key idea is that the relationship between a body and its observed surroundings provides evidence for both the scene’s metric reference and the body’s placement within it. Human structure specifies where and how to query scene evidence; attention encodes the resulting associations into relation features, and geometric readouts translate these features into corrections. This connects calibration and placement through a common formulation: encode human–scene interaction, then infer the geometry it supports. Interaction is represented by these learned relations and their induced updates, without an intermediate contact-label prediction.

Body-Anchored Metric Calibration first establishes the reference in which regional relations can be compared. Human-surface depth correspondences provide an analytic scale initialization. Our *Body-Anchored Scene Query* then relates geometrically conditioned body anchors to local scene features, forming a metric interaction representation. A metric readout predicts residual scale and depth corrections, and aggregation across people and frames yields a shared calibration for scene depth and camera translation. Human geometry therefore supplies both the metric prior and the conditions under which scene evidence is interpreted.

Dynamic Interaction Grounding encodes the relation between the current body and the calibrated scene. Our *Hierarchical Context Attention* combines body-relative geometry across local and broader supports, treating self-surface and environmental observations as complementary evidence. The resulting regional interaction representations support a placement readout that predicts and aggregates whole-body translation proposals according to their reliability. Updating the body changes these relations: we reproject it, rebuild the context, and encode interaction again. This dynamic loop refines root translation while preserving pose and shape, learning how human–scene relations should influence placement without explicitly classifying contact.

Our contributions are summarized as follows:

- We introduce **GroundedHuman**, a framework for monocular dynamic human–scene reconstruction that infers metric calibration and human placement from implicit interaction representations. This relation-based formulation reconstructs humans, scene geometry, and cameras in a shared metric world without explicit contact prediction.
- We propose **Body-Anchored Metric Calibration**, whose *Body-Anchored Scene Query* encodes human–scene correspondence for a residual metric readout. Together with surface-based initialization and temporal aggregation, it establishes a common metric reference for scene depth and camera translation.

- We develop **Dynamic Interaction Grounding**, whose *Hierarchical Context Attention* forms regional interaction representations for reliable translation updates. Re-encoding the relations at each updated body location enables dynamic placement refinement while preserving pose and shape.

2 RELATED WORK

3D Scene Reconstruction Recent visual geometry foundation models have enabled direct prediction of scene geometry and camera motion from RGB observations. DUST3R estimates pixel-aligned 3D pointmaps, while MAST3R introduces dense feature matching to establish cross-view correspondences (Wang et al., 2024a; Leroy et al., 2024). VGGT jointly predicts camera parameters, depth, and dense geometry, whereas π^3 achieves reference-free reconstruction through permutation-equivariant geometric prediction (Wang et al., 2025a;c). Extending reconstruction to dynamic and sequential observations, MonST3R models geometry in the presence of motion, CUT3R maintains a persistent state for continuous scene reconstruction, and VGGT- Ω advances feed-forward geometry estimation for both static and dynamic scenes (Zhang et al., 2024; Wang et al., 2025b; 2026a). Despite these advances, monocular scene reconstruction remains subject to scale ambiguity, while moving people introduce occlusion and depth discontinuities. Consequently, scene geometry and independently reconstructed human bodies may lack a consistent metric scale, leading to inaccurate spatial alignment.

Human Reconstruction and Global Motion Human Mesh Recovery (HMR) estimates 3D human pose and shape, commonly using parametric body models such as SMPL and SMPL-X (Loper et al., 2023; Pavlakos et al., 2019). HMR 2.0 and Multi-HMR recover expressive human meshes from monocular images, while NLF employs a continuous representation for dense human pose and shape estimation (Goel et al., 2023; Baradel et al., 2024; Sárádi & Pons-Moll, 2024). However, accurate local body reconstruction does not guarantee correct global placement. World-grounded HMR methods further recover human motion in a shared world coordinate frame. GLAMR uses learned motion priors to recover global trajectories under occlusion, while SLAHMR combines relative camera motion with human motion priors for global reconstruction (Yuan et al., 2022; Ye et al., 2023). TRAM leverages metric depth for camera calibration, WHAM combines motion context with contact-aware trajectory refinement, and GVHMR introduces gravity-view coordinates for world-grounded motion recovery (Wang et al., 2024b; Shin et al., 2024; Shen et al., 2024). More recently, DuoMo separates camera-space and world-space motion generation into two diffusion stages (Wang et al., 2026b). Nevertheless, temporally coherent human motion does not ensure accurate placement within the surrounding scene, as global localization errors can persist without sufficient constraints from scene geometry.

Scene-Human Reconstruction Recent human–scene reconstruction methods increasingly exploit human priors and scene geometry for joint reconstruction. SynCHMR uses human-aware metric SLAM to recover metric camera poses and scene geometry, followed by scene-aware human motion refinement (Zhao et al., 2024). MetricHMSR incorporates camera-ray representations for metric human recovery and uses reconstructed human surfaces as geometric anchors to refine monocular scene depth (Song et al., 2026). JOSH jointly optimizes camera poses, scene geometry, and human motion through correspondence and contact constraints (Liu et al., 2026). These approaches establish metric consistency through human priors or joint optimization, while recent feed-forward methods integrate human and scene reasoning directly into geometric prediction. HAMSt3R augments multi-view reconstruction with human segmentation and dense body correspondences, while Human3R incorporates human prompts into a recurrent geometry model to jointly recover scene structure, camera motion, and global human meshes (Rojas et al., 2025; Chen et al., 2026). UniSH further combines scene reconstruction and HMR priors with explicit human–scene geometric correspondence for joint metric-scale reconstruction (Li et al., 2026). However, metric-scale reconstruction and joint geometric prediction do not fully resolve human–scene interpenetration and floating artifacts, particularly under occlusion and ambiguous depth. To improve physical consistency, Uni-Con3R feeds predicted human–scene contact back into human reconstruction as a corrective signal (Sur et al., 2026), while GRAFT employs body-anchored geometric probes and recurrent refinement to correct human meshes according to nearby scene surfaces (Pradyumna YM et al., 2026). Although geometric feedback improves human–scene alignment, jointly refining body parameters may introduce unnecessary changes to local articulation when correcting global placement errors. This highlights the challenge of improving human–scene consistency while preserving reliable local body pose and shape.

Method overview to be redrawn.

Body-Anchored Metric Calibration $\rightarrow$ Dynamic Interaction Grounding

Figure 2: GroundedHuman infers metric calibration and human placement from implicit interaction representations. Body-Anchored Scene Query encodes human–scene correspondence, whose metric readout calibrates scene depth and camera translation. Within this reference, Hierarchical Context Attention encodes regional relations, whose placement readout updates whole-body translation. Updating the body changes its relation to the scene, so the regional representation is recomputed before the next refinement step.

3 METHOD

Given a monocular video $\mathcal{I} = \{\mathbf{I}_t\}_{t=1}^T$, GroundedHuman recovers scene geometry, camera motion, and human meshes in a shared metric world coordinate system. We formulate human–scene grounding as learning *implicit interaction representations*, from which metric calibration and human placement are inferred. These representations encode how scene observations relate to a body’s metric extent and current spatial configuration. Human structure determines where and how to query the scene; attention integrates the relevant evidence into a relational representation, and geometric readouts convert it into corrections. The organizing principle is

$$(\mathcal{H}, \mathcal{S}) \xrightarrow{\text{Query / Attention}} \mathbf{Z}_{\text{int}} \xrightarrow{\text{Geometric readout}} (\text{Metric, Placement}),$$

where $\mathcal{H}$ and $\mathcal{S}$ denote human and scene observations. Interaction is implicit in the learned relation features $\mathbf{Z}_{\text{int}}$ and the corrections they support, without an intermediate contact-label prediction.

We instantiate this principle with two successive modules and their respective representations and readouts (Fig. 2). **Body-Anchored Metric Calibration** uses *Body-Anchored Scene Query* to encode human–scene correspondence and read out residual metric corrections. **Dynamic Interaction Grounding** uses *Hierarchical Context Attention* to encode regional human–environment relations and read out placement updates. Their ordering addresses an ambiguity: the same local depth discrepancy may reflect a scene-scale error or a misplaced human. Establishing the common metric reference first makes regional relations comparable for placement reasoning. Each placement update then changes the relations themselves, prompting the interaction representation to be recomputed.

Initial observations. Frozen VGGT- Ω provides relative Z-depth D_t^r , intrinsics $\mathbf{K}_t$, extrinsics $(\mathbf{R}_{cw,t}, \mathbf{t}_{cw,t}^r)$, and multi-layer scene features $\mathbf{F}_t$. Under the same intrinsics, frozen NLF provides metric SMPL observations: pose θ_{tq} , shape β_{tq} , root translation $\tau_{tq}^{(0)}$, and confidence c_{tq} for person q . Let $\bar{\mathbf{V}}_{tq}$ denote the posed vertices before root translation; the current camera-space body is $\mathbf{V}_{tq}^{c,(k)} = \bar{\mathbf{V}}_{tq} + \tau_{tq}^{(k)}$, with translation broadcast over vertices. These estimates supply the human metric prior; pose and shape are preserved throughout both modules.

3.1 BODY-ANCHORED METRIC CALIBRATION

The first module encodes the correspondence between metric human observations and relative scene geometry. Surface depth ratios establish a coarse reference; body-anchored queries then form a relation representation from which the remaining metric correction is predicted.

Analytic Scale Initialization. Projecting an initial body vertex into D_t^r yields a pair of depths along the same camera ray. We pool valid depth ratios across people:

$$s_t^c = \text{median}_{(q,i) \in \mathcal{A}_t} \frac{[\mathbf{V}_{tqi}^{c,(0)}]_z}{D_t^r \left(\pi(\mathbf{K}_t, \mathbf{V}_{tqi}^{c,(0)}) \right)}, \quad D_t^{\text{coarse}} = s_t^c D_t^r. \quad (1)$$

Here, π is perspective projection and $\mathcal{A}_t$ retains valid correspondences after projection and depth filtering, keeping the nearest sampled human surface when projections coincide. The median reduces sensitivity to isolated errors. Sampling and missing-observation handling are detailed in Appendix C.

Body-Anchored Scene Query. An anchor specifies both *where* to retrieve scene evidence and *how* to interpret it relative to the body. We construct 24 anchor tokens per person from the body state, anchor position and projection, and a nearby observation from D_t^{coarse} : its 3D position, normal, anchor-relative offset, distance, and depth residual. A 3×3 neighborhood of $\mathbf{F}_t$ supplies scene tokens $\mathbf{S}_{tqj}$ for anchor j . Self-attention relates evidence across the body, while cross-attention associates each anchor with its local scene context. Let $\mathbf{H}_{tq}^{(l)}$ denote the anchor tokens at layer l :

$$\tilde{\mathbf{H}}_{tq}^{(l)} = \text{SelfAttn}(\mathbf{H}_{tq}^{(l)}), \quad \mathbf{h}_{tqj}^{(l+1)} = \text{CrossAttn}(\tilde{\mathbf{h}}_{tqj}^{(l)}, \mathbf{S}_{tqj}). \quad (2)$$

In cross-attention, body anchors supply the queries and scene tokens supply the keys and values; normalization, residual connections, and feed-forward blocks are omitted for clarity. The resulting tokens encode body-conditioned scene evidence. Pooling the final responses gives the metric interaction representation $\mathbf{z}_{tq}^{\text{met}} = \text{Pool}(\mathbf{H}_{tq}^{(L)})$. A metric readout maps this representation to residual log-scale $\Delta \ell_{tq}$ and depth bias Δb_{tq} . Thus geometric correction is inferred from the learned human–scene relation, with analytic initialization accounting for the dominant scale mismatch.

Temporal Aggregation. Normalized NLF confidences combine person-level predictions into frame-level residual scale ρ_t and bias b_t (Appendix C). Assuming a common scene scale within the input clip, we aggregate

$$\begin{aligned} s^{\text{vid}} &= \exp \left[\text{median}_t \log(s_t^c \rho_t) \right], & b^{\text{vid}} &= \text{median}_t b_t, \\ D_t^m &= s^{\text{vid}} D_t^r + b^{\text{vid}}, & \mathbf{t}_{cw,t}^m &= s^{\text{vid}} \mathbf{t}_{cw,t}^r. \end{aligned} \quad (3)$$

Only depth receives the additive bias; camera rotation and intrinsics are unchanged. Back-projecting D_t^m with $\mathbf{K}_t$ gives camera-space metric points $\mathbf{P}_t^m$, the reference for subsequent placement refinement.

3.2 DYNAMIC INTERACTION GROUNDING

With the metric reference established, the second module encodes how a body is situated relative to its surroundings and reads out a translation correction. Moving the body changes the relevant observations and geometric residuals. Its interaction representation must therefore depend on the current placement and be recomputed as that placement is refined.

Hierarchical Context Attention. We partition the SMPL surface into 96 corresponding regions, each represented by its centroid and eight sampled vertices. At iteration k , each region gathers scene points from four supports, $\mathcal{W} = \{\mathcal{W}_3, \mathcal{W}_7, \mathcal{W}_{\text{adp}}, \mathcal{W}_{\text{ann}}\}$: fixed 3×3 and 7×7 windows, an adaptive window following its projected extent, and an outer annulus. These supports expose fine alignment cues together with broader environmental structure.

Each sample is encoded relative to the current body region using 3D and image-plane offsets and depth residuals. Comparison with an instance-aware human depth buffer separates observations associated with the person’s own surface from environmental evidence. Other-human samples are excluded from both attention channels and retained only as interference statistics. Masked attention selects and pools the relation features within each support and channel, yielding eight context tokens per region together with valid-support ratios. Empty supports return zero tokens. Fusing this context with region geometry, body-region embeddings, and interference statistics gives a regional interaction representation $\mathbf{z}_r^{\text{place},(k)}$. It encodes the body’s agreement with observed human surfaces and its spatial relationship to the environment, supplying complementary evidence for placement. The pooling formulation is provided in Appendix C.

Interaction-Aware Placement. A placement readout maps $\mathbf{z}_r^{\text{place},(k)}$ to a bounded translation proposal $\mathbf{v}_r^{(k)}$, gate $g_r^{(k)}$, and scalar log-variance $\ell_r^{(k)}$. This connects the implicit interaction representation to a concrete geometric action. Proposals are expressed along the region’s viewing direction

and two tangent directions, then converted to camera coordinates. Each proposes a translation of the *whole body*. With $m_r^{(k)}$ indicating valid support, we aggregate

$$\alpha_r^{(k)} = m_r^{(k)} g_r^{(k)} \exp(-\ell_r^{(k)}), \quad \hat{\mathbf{v}}^{(k)} = \frac{\sum_r \alpha_r^{(k)} \mathbf{v}_r^{(k)}}{\max(\sum_r \alpha_r^{(k)}, \varepsilon)}. \quad (4)$$

The readout favors supported, reliable evidence. Supervision on these corrections trains the interaction representation without requiring explicit contact labels.

Translation Refinement. A person-level gate modulates the aggregated proposal after component-wise clipping (Appendix C), producing the root update $\Delta\tau_{tq}^{(k)}$:

$$\tau_{tq}^{(k+1)} = \tau_{tq}^{(k)} + \Delta\tau_{tq}^{(k)}. \quad (5)$$

Two feed-forward steps share parameters. After each update, we rebuild region locations, sample ownership, and attention context to obtain a new $\mathbf{z}_r^{\text{place},(k+1)}$. This closes the loop between relation encoding and geometric readout: an update changes the human–scene relationship on which the next update depends. Here, *dynamic* refers to this placement-dependent recomputation. Scene geometry, camera parameters, body pose, and shape remain fixed. Finally, the calibrated inverse camera extrinsics map the refined bodies and metric scene points to a shared world frame (Appendix B).

3.3 TRAINING OBJECTIVES

The two stages are trained separately, with VGGT- Ω and NLF frozen. Metric calibration uses metrically supervised sequences with synthetic scale perturbations. Initial scale training is followed by training on analytically initialized states, supervising the residual correction required from that state:

$$\mathcal{L}_{\text{calib}} = \lambda_s \mathcal{L}_{\text{scale}} + \lambda_b \mathcal{L}_{\text{bias}} + \lambda_d \mathcal{L}_{\text{depth}}. \quad (6)$$

The terms supervise residual log-scale, depth bias, and calibrated depth. In the residual-head training stage, the anchor interaction backbone is frozen and only the scale and bias heads are updated. The bias target is zero for this scale-perturbation path, constraining additive compensation for multiplicative errors.

With calibration fixed, Dynamic Interaction Grounding is trained on clean, scale-perturbed, translation-perturbed, and mixed states:

$$\mathcal{L}_{\text{ground}} = \lambda_v \mathcal{L}_{\text{vote}} + \lambda_t \mathcal{L}_{\text{trans}} + \lambda_g \mathcal{L}_{\text{gate}} + \lambda_r \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{state}}. \quad (7)$$

Regional votes target the remaining displacement divided by the number of remaining updates. Final-translation supervision trains the placement readout, gate supervision trains update validity, and regularization controls update magnitude. State-dependent terms encourage recovery from large errors, preservation of accurate inputs, and non-degrading corrections across iterations. Appendix D details the objectives and training configuration status.

4 EXPERIMENTS

We evaluate global human motion on EMDB-2 and RICH, scene depth on Bonn, camera trajectories on TUM-Dynamic, and local human meshes on 3DPW and EMDB-1. Qualitative comparisons on PROX-D and outdoor sequences examine human–scene alignment, while measurements on PROX assess human and environmental dimensions. Implementation details are provided in Appendix D.

4.1 GLOBAL HUMAN MOTION ESTIMATION

We assess global human motion on EMDB-2 and RICH using WA-MPJPE, W-MPJPE, and RTE. The joint metrics assess world-space motion under different alignment scopes, while RTE measures root-trajectory accuracy.

Category	Method	Preprocessed Input (✓ = not required)			Output			EMDB-2 (24)			RICH (24)		
		Tracking	Camera	Depth	Multi-Person	Metric Human-Scene	Contact	WA-MPJPE ↓	W-MPJPE ↓	RTE ↓	WA-MPJPE ↓	W-MPJPE ↓	RTE ↓
Offline	GLAMR	×	✓	✓	×	×	×	280.8	726.6	11.4	129.4	236.2	3.8
	SLAHMR	×	×	✓	✓	×	×	326.9	776.1	10.2	132.2	237.1	6.4
	COIN	×	×	✓	×	×	×	152.8	407.3	3.5	169.5	254.5	–
	GVHMR	×	×	✓	×	×	×	111.0	276.5	2.0	78.8	126.3	2.4
	TRAM	×	×	×	×	×	×	76.4	222.4	1.4	127.8	238.0	6.0
	JOSH	×	×	×	×	×	✓	68.9	174.7	1.3	89.0	132.5	3.0
	PromptHMR	✓	✓	✓	×	×	×	71.0	216.5	1.3	118.2	207.0	5.6
Online	TRACE	✓	✓	✓	✓	×	×	529.0	1702.3	17.7	238.1	925.4	610.4
	WHAM	×	×	✓	×	×	×	135.6	354.8	6.0	108.4	196.1	4.5
	JOSH3R	×	×	✓	×	×	×	220.0	661.7	13.1	–	–	–
	Human3R	✓	✓	✓	✓	✓	×	112.2	267.9	2.2	110.0	184.9	3.3
	MetricHMSR	×	×	×	×	×	×	72.1	199.5	1.4	109.6	165.5	1.9
	UniCon3R	×	×	×	×	×	✓	113.7	285.6	2.3	81.5	129.8	1.7
	UniSH	×	×	×	×	×	✓	118.5	270.1	5.8	118.1	183.2	4.8
	Ours	✓	✓	✓	✓	✓	✓	58.5	150.0	1.1	87.7	128.2	2.5

Table 1: Evaluation of global human motion estimation on EMDB-2 and RICH.

World-space human motion depends on both camera motion and scene scale. Our comparison includes dedicated motion estimators and joint reconstruction methods (Table 1). GroundedHuman places the body sequence within its environment through a scale shared by scene depth and camera translation.

The reconstructed sequences expose articulation and global displacement within the same scene (Fig. 3). Sports warm-up, handstand, and walking-with-lunges examples show bodies at successive instants, alongside predicted and reference world trajectories. This joint view relates each motion path to the surrounding geometry rather than presenting it as an isolated trajectory.

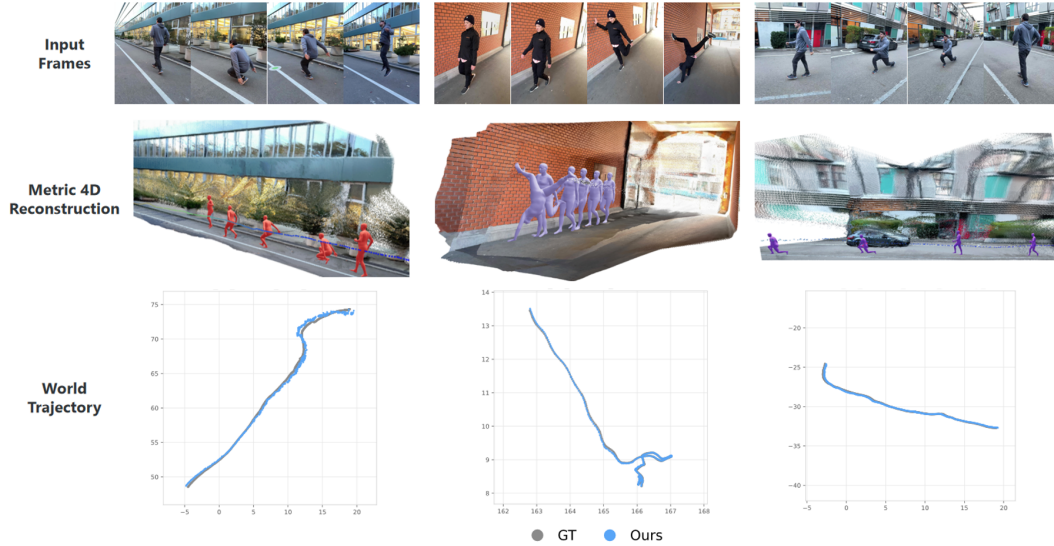


Figure 3: Temporal human–scene reconstruction. Each column shows input frames, reconstructed bodies within the scene, and world trajectories. Gray: ground truth; blue: ours.

4.2 SCENE RECONSTRUCTION

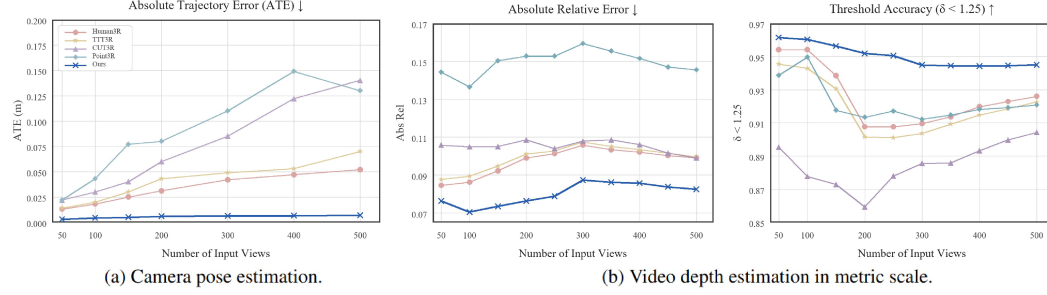


Figure 4: Evaluation of generic 3D reconstruction across different input lengths. (a) Camera pose estimation on TUM-Dynamic using absolute trajectory error (ATE) after Sim(3) trajectory alignment. (b) Video depth estimation on Bonn in metric scale using Abs Rel and $\delta < 1.25$. For Grounded-Human, metric depth is obtained with the Body-Anchored Metric Calibration module in Sec. 3.1, without per-sequence ground-truth scale or shift fitting.

Camera Pose Estimation. Following prior work, Fig. 4(a) evaluates camera trajectories on TUM-Dynamic using ATE after Sim(3) alignment. The comparison covers different input lengths and representative online reconstruction models. GroundedHuman retains the backbone’s camera rotations and applies the video-level scale recovered by Body-Anchored Metric Calibration to camera translation, producing stable trajectory accuracy as the input sequence grows.

Video Depth Estimation. Figure 4(b) evaluates video depth on Bonn using Abs Rel and $\delta < 1.25$, which measure relative depth error and the fraction of predictions within a factor of 1.25 of the ground truth, respectively. The predicted relative depth is converted to metric scale by the Body-Anchored Metric Calibration module in Sec. 3.1: human surfaces initialize scale, body-anchored scene queries predict residual scale and depth bias, and temporal aggregation applies one shared calibration across the video. No per-sequence ground-truth scale or shift is fitted for these depth results. Across input lengths, the resulting metric depth remains more accurate than the compared online baselines on both measures.

4.3 HUMAN MESH RECOVERY

We evaluate local human mesh recovery on 3DPW and EMDB-1 using PA-MPJPE, MPJPE, and PVE, measured in millimeters. Joint errors assess articulated body geometry, while vertex error additionally captures surface reconstruction. The comparison includes dedicated human mesh estimators and joint reconstruction methods (Table 2).

Method	Crop-free	Detection-free	Intrinsic-free	3DPW (14)			EMDB-1 (24)		
				PA-MPJPE ↓	MPJPE ↓	PVE ↓	PA-MPJPE ↓	MPJPE ↓	PVE ↓
CLIFF	×	×	×	43.0	69.0	81.2	68.3	103.3	123.7
HMR2.0	×	×	✓	44.4	69.8	82.2	61.5	97.8	120.0
ReFit	×	×	×	—	—	—	58.6	88.0	104.5
TokenHMR	×	×	✓	44.3	71.0	84.6	55.6	91.7	109.4
CameraHMR	×	×	✓	38.5	62.1	72.9	43.7	73.0	85.4
BEV	✓	✓	✓	46.9	78.5	92.3	70.9	112.2	133.4
Multi-HMR	✓	✓	×	45.9	73.1	87.1	50.1	81.6	95.7
Human3R	✓	✓	✓	44.1	71.2	84.9	48.5	73.9	86.0
Ours	×	×	✓	37.3	60.3	71.4	41.2	69.6	82.4

Table 2: Evaluation of local human mesh recovery on 3DPW and EMDB-1.

GroundedHuman retains the local pose and shape supplied by the frozen NLF observer. Scene calibration and root-translation refinement operate on global geometry, leaving the articulated body representation unchanged. Local mesh evaluation therefore complements the world-motion comparison: together, they distinguish the quality of the recovered body from its placement and movement within the environment.

4.4 ABLATION STUDY

Following the physical-grounding metrics reported by UniCon3R on RICH (Sur et al., 2026), we compare the shared VGGT–NLF–HSI scale stage with its contact-aware body-region-query variant. Table 3 reports collision ratio (Coll.), penetration (Pen.), floating distance (Float), and maximum penetration (P.Max) for our two variants. Adding body-region queries lowers Coll. from 12.83% to 6.48% (6.35 percentage points), Pen. from 4.65 to 2.89 cm, Float from 17.15 to 13.47 cm, and P.Max from 7.99 to 7.31 cm. The smaller change in P.Max suggests that the largest penetration errors remain harder to correct than the aggregate measures.

Variant	Coll. (%) ↓	Pen. (cm) ↓	Float (cm) ↓	P.Max (cm) ↓
VGGT–NLF–HSI scale	12.83	4.65	17.15	7.99
+ body-region query	6.48	2.89	13.47	7.31

Table 3: Physical-grounding ablation using the four diagnostics reported by UniCon3R (Sur et al., 2026). Both rows are our variants; lower is better.

4.5 QUALITATIVE RESULTS

Human–Scene Spatial Alignment. We compare human placement with UniSH and Human3R on PROX-D and outdoor sequences containing lying, reclining, and sitting poses (Fig. 5). The indoor scenes emphasize torso-to-cushion and pelvis-to-seat alignment, while the outdoor scenes expose the position of the body relative to benches and backrests. These relationships reveal global offsets that can remain inconspicuous when a body is inspected in isolation. Our regional refinement uses the surrounding geometry to adjust root position while preserving the observer’s local articulation.

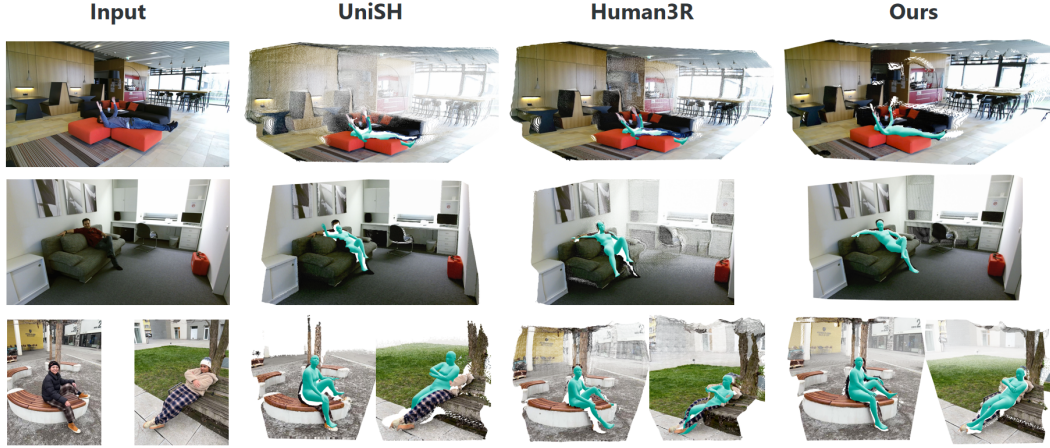


Figure 5: Human placement in reconstructed scenes. Columns show the input, UniSH, Human3R, and GroundedHuman. Lying, reclining, and outdoor sitting examples highlight the position of the body relative to nearby support surfaces.

Metric Measurements. On PROX, we compare reconstructed human height and furniture dimensions with reference measurements and MapAnything reconstructions (Fig. 6). The standing-person example places body height and table length in the same coordinate system, while the seated example relates the person to the surrounding partitions. These measurements complement the placement comparison: a body may appear aligned with nearby surfaces from one viewpoint even when the overall scene dimensions are inaccurate. Inspecting both human and environmental lengths therefore provides a physical interpretation of the joint geometry beyond its projected appearance. The examples also expose how scale affects objects at different positions within the scene, rather than only the human used as a reference.

Together, these evaluations connect local body geometry, global motion, and scene-relative placement within a single reconstruction.

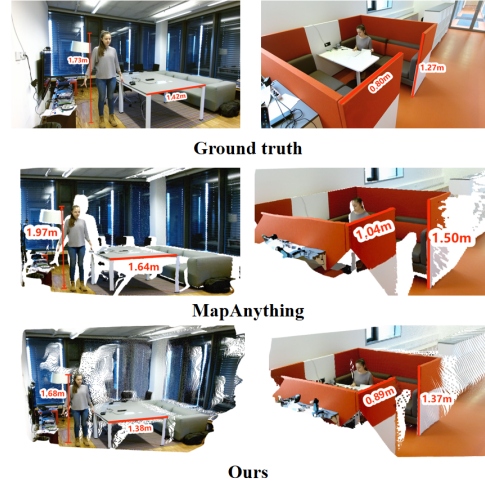


Figure 6: Comparison of the metric measurement on PROX.

5 CONCLUSION

We presented GroundedHuman, which formulates metric calibration and human placement as geometric readouts of implicit interaction representations. Body-Anchored Scene Query encodes human–scene correspondence to establish a shared metric reference; Hierarchical Context Attention encodes regional relations within that reference to infer placement corrections. Updating placement changes the relations themselves, so Dynamic Interaction Grounding recomputes its representation before the next correction. This connects queries, attention, and geometric updates through a common relation-based formulation while retaining the pose and shape supplied by pretrained estimators. Evaluations examine world-space motion, scene geometry, and human–scene spatial consistency. Extending these interaction representations to temporal evidence and articulated corrections is a direction for future work.

Limitations. GroundedHuman depends on the quality and visibility of the initial human estimates and on the accuracy of scene depth. Biased body shape or root position can affect scale calibration, while occlusion and local depth errors can weaken the evidence for translation refinement. Without

reliable human observations, the metric scale is weakly constrained. Preserving pose and shape prevents global alignment errors from being absorbed by local deformation, but also leaves incorrect articulation and body shape unchanged. Translation alone cannot resolve all human–scene contact errors, particularly when these require changes in pose or scene geometry.

ACKNOWLEDGMENTS

Omitted in the double-blind review version.

GENERATIVE AI USE DISCLOSURE

Generative AI tools were used to assist with the organization and language editing of the Chinese draft, its English translation, and early sketches of conceptual figures. All technical designs, code, experimental data, equations, literature claims, and final content were reviewed by the authors; the authors take full responsibility for the contents of this paper.

ETHICS STATEMENT

This work uses public human and scene research datasets for method development and evaluation, and does not target identity recognition or individual-attribute inference. Human-video reconstruction may raise privacy concerns, risks of surveillance misuse, and scale bias across body types; Appendix F discusses these boundaries. Dataset licenses and conditions of use will be checked individually before final submission, and no unverified information is inferred in this manuscript.

REPRODUCIBILITY STATEMENT

The appendix documents coordinate and scale conventions, the complete inference procedure, training objectives, datasets, and evaluation protocols, and explicitly marks ablation, robustness, and efficiency results that still require real experiments. Before final submission, all placeholders will be replaced with verified server configurations, anonymous code, execution scripts, and real experimental outputs. No quantitative conclusion in this paper is supported by placeholder content.

REFERENCES

- Fabien Baradel, Matthieu Armando, Salma Galaaoui, Romain Brégier, Philippe Weinzaepfel, Grégory Rogez, and Thomas Lucas. Multi-hmr: Multi-person whole-body human mesh recovery in a single shot. 2024.
- Yue Chen, Xingyu Chen, Yuxuan Xue, Anpei Chen, Yuliang Xiu, and Gerard Pons-Moll. Human3r: Everyone everywhere all at once. In *International Conference on Learning Representations*, 2026. arXiv:2510.06219.
- Shubham Goel, Georgios Pavlakos, Jathushan Rajasegaran, Angjoo Kanazawa, and Jitendra Malik. Humans in 4d: Reconstructing and tracking humans with transformers. 2023.
- Vincent Leroy, Yohann Cabon, and Jérôme Revaud. Grounding image matching in 3d with mast3r. In *European Conference on Computer Vision*, 2024.
- Mengfei Li, Peng Li, Zheng Zhang, Jiahao Lu, Chengfeng Zhao, Wei Xue, Qifeng Liu, Sida Peng, Wenxiao Zhang, Wenhan Luo, Yuan Liu, and Yike Guo. Unish: Unifying scene and human reconstruction in a feed-forward pass. *arXiv preprint arXiv:2601.01222*, 2026.
- Zhizheng Liu, Joe Lin, Wayne Wu, and Bolei Zhou. Joint optimization for 4d human-scene reconstruction in the wild. In *International Conference on Learning Representations*, 2026. arXiv:2501.02158.
- Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. Smpl: A skinned multi-person linear model. In *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, pp. 851–866. Association for Computing Machinery, 2023.

- Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive body capture: 3d hands, face, and body from a single image. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10975–10985, 2019.
- Pradyumna YM, Yuxuan Xue, Yue Chen, Nikita Kister, István Sárándi, and Gerard Pons-Moll. Graft: Geometric refinement and fitting transformer for human scene reconstruction. *arXiv preprint arXiv:2604.19624*, 2026.
- Sara Rojas, Matthieu Armando, Bernard Ghanem, Philippe Weinzaepfel, Vincent Leroy, and Grégoire Rogez. Hamst3r: Human-aware multi-view stereo 3d reconstruction. *arXiv preprint arXiv:2508.16433*, 2025.
- István Sárándi and Gerard Pons-Moll. Neural localizer fields for continuous 3d human pose and shape estimation. In *Advances in Neural Information Processing Systems*, 2024.
- Zehong Shen, Huaijin Pi, Yan Xia, Zhi Cen, Sida Peng, Zechen Hu, Hujun Bao, Ruizhen Hu, and Xiaowei Zhou. World-grounded human motion recovery via gravity-view coordinates. In *SIGGRAPH Asia Conference Papers*, 2024.
- Soyong Shin, Juyong Kim, Eni Halilaj, and Michael J. Black. Wham: Reconstructing world-grounded humans with accurate 3d motion. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2070–2080, 2024.
- Chentao Song, He Zhang, Haolei Yuan, Haozhe Lin, Jianhua Tao, Hongwen Zhang, and Tao Yu. Metrichmsr: Metric human mesh and scene recovery from monocular images. *arXiv preprint arXiv:2506.09919*, 2026.
- Tanuj Sur, Shashank Tripathi, Nikos Athanasiou, Ha Linh Nguyen, Kai Xu, Michael J. Black, and Angela Yao. Unicon3r: Unified contact-aware 4d human-scene reconstruction from monocular video. *arXiv preprint arXiv:2604.19923*, 2026.
- Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025a.
- Jianyuan Wang, Minghao Chen, Shangzhan Zhang, Nikita Karaev, Johannes Schönberger, Patrick Labatut, Piotr Bojanowski, David Novotny, Andrea Vedaldi, and Christian Rupprecht. Vggt- ω . *arXiv preprint arXiv:2605.15195*, 2026a.
- Qianqian Wang, Yifei Zhang, Aleksander Holynski, Alexei A. Efros, and Angjoo Kanazawa. Continuous 3d perception model with persistent state. *arXiv preprint*, 2025b.
- Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jérôme Revaud. Dust3r: Geometric 3d vision made easy. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024a.
- Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. π^3 : Permutation-equivariant visual geometry learning. *arXiv preprint arXiv:2507.13347*, 2025c.
- Yufu Wang, Ziyun Wang, Lingjie Liu, and Kostas Daniilidis. Tram: Global trajectory and motion of 3d humans from in-the-wild videos. In *European Conference on Computer Vision*, pp. 467–487, 2024b.
- Yufu Wang, Evonne Ng, Soyong Shin, Rawal Khirodkar, Yuan Dong, Zhaoen Su, Jinhyung Park, Kris Kitani, Alexander Richard, Fabian Prada, and Michael Zollhöfer. Duomo: Dual motion diffusion for world-space human reconstruction. *arXiv preprint arXiv:2603.03265*, 2026b.
- Vickie Ye, Georgios Pavlakos, Jitendra Malik, and Angjoo Kanazawa. Decoupling human and camera motion from videos in the wild. 2023.

- Ye Yuan, Umar Iqbal, Pavlo Molchanov, Kris Kitani, and Jan Kautz. Glamr: Global occlusion-aware human mesh recovery with dynamic cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11038–11049, 2022.
- Junyi Zhang, Charles Herrmann, Junhwa Hur, Varun Jampani, Trevor Darrell, Forrester Cole, Deqing Sun, and Ming-Hsuan Yang. Monst3r: A simple approach for estimating geometry in the presence of motion. *arXiv preprint arXiv:2410.03825*, 2024.
- Yizhou Zhao, Tuanfeng Yang Wang, Bhiksha Raj, Min Xu, Jimei Yang, and Chun-Hao Paul Huang. Synergistic global-space camera and human reconstruction from videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1216–1226, 2024.

A EVALUATION PROTOCOLS AND SUPPLEMENTARY RESULTS

This appendix provides evaluation protocols, results across sequence lengths, and additional visualizations for Sec. 4.

A.1 BONN DEPTH PROTOCOLS

The Bonn evaluation uses balloon2, crowd2, crowd3, person_tracking2, and synchronous. We evaluate metric depth without per-sequence ground-truth scale or shift fitting. GroundedHuman obtains metric scale from the Body-Anchored Metric Calibration module in Sec. 3.1. For sequence-length evaluation, each window begins at frame 30, and sequences shorter than the request contribute all remaining frames without padding. Metrics are aggregated using valid depth pixels across the five sequences.

Table 4 reports depth accuracy without GT alignment as the requested input length varies from 50 to 500 views. This evaluation complements the main comparison by examining the dependence of scene calibration on the temporal extent of the input.

Table 4: Bonn depth without GT scale or shift alignment across input lengths. Metrics are weighted by valid pixels across five sequences; actual frame counts account for shorter sequences.

Requested views	Actual frames	Abs Rel ↓	$\delta < 1.25$ ↑
50	50	0.17049	0.82854
100	100	0.14637	0.86971
150	150	0.14197	0.86684
200	200	0.13640	0.88192
250	250	0.15130	0.84632
300	300	0.18281	0.74750
350	302–350	0.20185	0.68086
400	302–400	0.21045	0.65423
450	302–450	0.20948	0.68589
500	302–500	0.20229	0.72605

A.2 CAMERA GEOMETRY AND METRIC-DEPTH DIAGNOSTICS

Table 5 reports camera-translation RMSE after Sim(3) alignment, averaged across eight TUM-Dynamic sequences. This alignment convention follows Human3R (Chen et al., 2026). Alignment removes the global similarity transform, making this evaluation a measure of relative camera-trajectory geometry.

Table 5: Backbone camera errors on TUM-Dynamic after Sim(3) alignment. Each value is the mean ATE over eight sequences.

Views	ATE (m)
50	0.00299
100	0.00434
150	0.00493
200	0.00591
300	0.00634
400	0.00645
500	0.00692

Figure 4 in Sec. 4.2 reports (a) Sim(3)-aligned camera ATE on TUM-Dynamic and (b) metric-depth accuracy on Bonn. The reference curves follow the sequence-length evaluation presented by Human3R. GroundedHuman’s depth curves use the scale and depth-bias correction predicted by Body-Anchored Metric Calibration rather than a scale or shift fitted from ground truth.

B NOTATION, COORDINATE SYSTEMS, AND METRIC GAUGE

Appendix B specifies coordinate and scale conventions; Appendix C gives the inference procedure; Appendix D summarizes training; Appendix E details the evaluation protocols; and Appendix F discusses limitations.

B.1 COORDINATE AND DEPTH CONVENTIONS

For frame t and person q in a clip, VGGT- Ω predicts relative scene depth and camera parameters, while NLF provides SMPL human observations expressed in meters. We establish correspondences between human surfaces and scene depth using Z-depth along the camera optical axis; Euclidean distance along a camera ray is not treated as Z-depth. Extrinsics follow the camera-from-world convention, $\mathbf{X}^c = \mathbf{R}_{cw,t} \mathbf{X}^w + \mathbf{t}_{cw,t}^m$, with points written as column vectors. Shared-scale depth and camera translation are defined in Eq. 3 and restated below; the inverse world transform follows directly from this camera-from-world convention. In the mesh equations, translation is broadcast over vertices.

Let D^{rel} denote relative depth, s_{clip} the shared clip-level scale, and b_{clip} the residual depth bias. Metric depth and metric camera translation are

$$D^{\text{metric}} = s_{\text{clip}} D^{\text{rel}} + b_{\text{clip}}, \quad \mathbf{t}^{\text{metric}} = s_{\text{clip}} \mathbf{t}^{\text{rel}}. \quad (8)$$

The bias acts only in the depth domain and is not directly added to camera translation. After shared-scale propagation, TRSTR performs a constrained camera-space translation update on the human root, while pose and shape remain frozen.

With $\bar{\mathbf{V}}_{tq}$ denoting the posed vertices before root translation, the refined human mesh is transformed vertexwise as

$$\mathbf{V}_{tq}^w = \mathbf{R}_{cw,t}^T \left(\bar{\mathbf{V}}_{tq} + \tau_{tq}^{(2)} - \mathbf{t}_{cw,t}^m \right). \quad (9)$$

The same inverse camera transform maps the metric scene points to the shared world frame.

B.2 TENSOR, PRECISION, AND DEVICE CONVENTIONS

Table 6: Main tensors and implementation conventions. Exact dimensions follow the final implementation configuration.

Object	Main dimensions	Unit/coordinates	Status
Input images	$B \times T \times 3 \times H \times W$	image plane	defined
Relative depth	$B \times T \times H \times W$	relative Z-depth	defined
SMPL vertices	$B \times T \times N \times V \times 3$	camera, meter	defined
Human anchors	$B \times T \times N \times 24 \times C$	feature space	defined
TRSTR regions	$B \times T \times N \times 96 \times C_r$	canonical body	defined
Exact dtype/device	–	–	To be evaluated

C MODEL ARCHITECTURE DETAILS AND COMPLETE INFERENCE

The residual calibration implementation is referred to as HSI in the configuration and architecture records below; its anchor–scene read implements Body-Anchored Scene Query in Sec. 3.1. TRSTR denotes the regional translation refiner underlying Dynamic Interaction Grounding in Sec. 3.2. These implementation names identify components of the two main modules, rather than additional stages. Hierarchical Context Attention uses the masked regional pooling described below to form the place–ment interaction representation; the dynamic update recomputes this representation at the revised body location.

C.1 VGGT- Ω , NLF, AND BASELINE BOUNDARIES

VGGT- Ω provides relative scene structure, camera parameters, and multi-level scene features. NLF provides local human pose, shape, camera-space translation, and confidence. GroundedHuman does

not replace either baseline, and TRSTR is not allowed to reduce world-space error by changing pose or shape. The final version must list frozen parameters, trainable parameters, input resolution, clip length, and the maximum number of person slots.

Placeholder: The frozen-module inventory, checkpoint paths, input shapes, dtypes, devices, and memory configuration must be verified against the server implementation.

C.2 SURFACE SAMPLING AND SCALE INITIALIZATION

Sampled vertices from all accepted people in a frame are pooled before projection. Pixel coordinates are rounded to the nearest integer, and only finite vertices with positive Z-depth and in-image projections are retained. The scene depth at each retained pixel must also be finite and positive. When several sampled vertices share a pixel, the nearest human surface sample is kept. Ratios outside the configured positive interval are removed, and the frame estimate is accepted only if enough pixels remain. Equation 1 takes a median over these surface ratios; it does not first average a scale independently predicted for each person.

The inspected spatial inference launcher uses vertex stride 8, at least 32 retained pixels, and ratio limits $[0.10, 10.0]$. These are inference defaults rather than a claim about every benchmark run. The accepted scale-training configuration uses a different disturbance range and validity policy. The nearest-sample test resolves conflicts among projected humans; it does not establish visibility with respect to an independently known environment. Consequently, systematic human-translation errors and environmental occlusion can still bias the scale.

Placeholder: The person-confidence threshold and all benchmark-specific sampling overrides must be checked against the evaluation records. Do not substitute training thresholds for inference settings.

C.3 PERSON-CONDITIONED RESIDUAL AFFINE

The 24 anchors comprise 21 non-root SMPL joints, two hand anchors obtained from adjacent wrist/hand joints, and a global anchor formed by averaging 27 fixed template vertices. Each token contains the body’s 6D joint rotations, shape coefficients and root translation, the anchor’s 3D position and normalized image projection, and geometric attributes sampled from the initialized depth D_t^c . The local probe searches a 9×9 window for the nearest valid scene point in 3D; when none is available, it uses the center depth sample. The attributes include the sampled point, surface normal, 3D offset, distance, and Z-depth residual.

Scene features from the final four cached aggregator layers are projected and concatenated into the scene representation. The spatial configuration uses 512D anchor tokens, eight attention heads, five interaction layers, and three passes through those layers. Each anchor attends to a 3×3 neighborhood of scene tokens around its projection. The token mean within a person is mapped to residual log-scale and bias. Confidence aggregation is

$$w_{tq} = \frac{\max(c_{tq}, 0)}{\max(\sum_j \max(c_{tj}, 0), \varepsilon)}, \quad (10)$$

$$\rho_t = \exp \left[\text{clip} \left(\sum_q w_{tq} \Delta \ell_{tq}, \ell_{\min}, \ell_{\max} \right) \right], \quad b_t = \sum_q w_{tq} \Delta b_{tq}.$$

The inspected configuration uses $\ell_{\min} = -5$ and $\ell_{\max} = 5$. If all person confidences are zero, the aggregation returns $\rho_t = 1$ and $b_t = 0$. This is an identity residual, not evidence that the scene’s absolute scale is known.

C.4 CLIP-LEVEL CONSENSUS AND CAMERA PROPAGATION

If a frame has no accepted analytic estimate, the sequence-median fallback substitutes the log-median scale of frames with successful surface estimates. Residual prediction then operates on this initialized depth. In the shared-world inference path, the composed scale $s_t^c \rho_t$ and the predicted bias are aggregated over the selected clip before TRSTR is run. The inspected wrapper takes the median over all selected frames after fallback; it does not apply an additional confidence mask to the final

temporal aggregation. Thus the temporal median in Eq. 3 includes frames whose initialization was filled from the clip, not only frames with directly observed surface support.

If no frame has a valid analytic scale, the wrapper emits a warning and retains its unit initialization. Such a clip has no supported human-derived metric scale; this numerical fallback must not be presented as successful absolute-scale recovery. The separate per-frame inference option does not apply the temporal aggregation in Eq. 3. Benchmark reports must identify which option was used.

C.5 COMPLETE TRSTR UPDATE

TRSTR assigns template vertices to anatomical groups using their dominant linear-blend-skinning joint, then partitions each group using deterministic farthest-point seeds. The spatial configuration contains 96 regions and eight representative vertices per region. Region identity is fixed; centers and representative points follow the current body geometry. Four support domains are used: 3×3 and 7×7 windows, an adaptive neighborhood, and an outer annulus. The adaptive radius is bounded by eight pixels and the annulus width is two pixels in the inspected configuration.

The human depth buffer stores the nearest projected human and its instance identity. A scene sample is associated with that human when the depth difference lies within the configured tolerance. Self-surface and environment observations are pooled separately in each support domain, producing eight tokens. Other-human samples are excluded from both pooling channels and retained as an interference ratio. Region validity requires a positive center depth, a valid person, and sufficient support in at least one pooled channel. The spatial configuration uses a minimum support ratio of 0.25, 16D probe tokens, a 32D anatomical embedding, and a 256D regional representation.

Let $\mathbf{e}_{ri}^{(k)}$ encode sample i relative to body region r at iteration k , and let $\mathcal{N}_{rsc}^{(k)}$ denote valid samples in support $s \in \{\mathcal{W}_3, \mathcal{W}_7, \mathcal{W}_{\text{adp}}, \mathcal{W}_{\text{ann}}\}$ and channel $c \in \{\text{self}, \text{env}\}$. Masked attention pooling is

$$\mathbf{u}_{rsc}^{(k)} = \sum_{i \in \mathcal{N}_{rsc}^{(k)}} \frac{\exp(a(\mathbf{e}_{ri}^{(k)}))}{\sum_{j \in \mathcal{N}_{rsc}^{(k)}} \exp(a(\mathbf{e}_{rj}^{(k)}))} \mathbf{e}_{ri}^{(k)}, \quad (11)$$

where a is a learned scalar score. Empty sample sets return zero tokens. The eight pooled tokens are fused with region geometry, anatomical embeddings, support ratios, and interference statistics to form $\mathbf{z}_r^{\text{place},(k)}$. Both the sample relations and their valid sets change with body placement; the attention is conditioned on this geometry through its inputs and sampling support.

$$\alpha_r = m_r g_r \exp(-\ell_r), \quad \hat{\mathbf{v}} = \frac{\sum_r \alpha_r \mathbf{v}_r}{\max(\sum_r \alpha_r, 10^{-5})}, \quad (12)$$

$$\Delta \tau = g^{\text{person}} \text{clip}_{[-d_{\max}, d_{\max}]^3}(\hat{\mathbf{v}}).$$

Each regional vote is predicted in the ray/tangent basis using bounded coefficients. Log-variance is scalar and clamped to $[-4, 4]$; it is not a directional covariance estimate. Equation 12 applies the component-wise camera-space clip before the person gate. Two iterations share parameters and re-evaluate region locations, sample ownership, and geometric support after each update. The spatial configuration disables the optional temporal branch.

Placeholder: Translation bounds differ between the inspected inference configuration and the later refinement-training configuration. Report the effective checkpoint and launch-time values for each experiment rather than assuming a single set of bounds.

Architecture details to be redrawn.

Figure 7: **Interaction representations and geometric readouts.** (a) Human-surface depth ratios initialize scene scale; Body-Anchored Scene Query relates 24 body anchors to local scene features, forming a metric interaction representation for residual scale and bias prediction. (b) Hierarchical Context Attention pools self-surface and environmental evidence across four supports around 96 body regions, forming regional interaction representations for whole-body translation proposals. Reprojecting and resampling after an update recomputes the relations for the next placement readout. Pose and shape are preserved.

C.6 INFERENCE ALGORITHM

1. Run the frozen VGGT- Ω to obtain relative depth, cameras, and scene features.
2. Run the frozen NLF to obtain local humans and confidence values.
3. Project the visible SMPL front surface and compute person/frame scale proposals.
4. Apply robust aggregation to obtain the analytic coarse scale.
5. Run the person-conditioned residual affine module to obtain residual scale and depth bias.
6. Aggregate the shared scale within the clip and propagate it to depth and camera translation.
7. Run two TRSTR iterations, updating only each person’s root translation.
8. Output humans, scene geometry, and camera trajectories in a shared world coordinate system.

D IMPLEMENTATION AND TRAINING DETAILS

D.1 STAGED TRAINING

Scale calibration and TRSTR are trained separately. The scale stage learns residual affine parameters from metric depth and SMPL supervision; the translation stage learns root corrections from perturbed human–scene states. VGGT- Ω and the human observer are not jointly optimized with these objectives. At inference, neither module receives ground-truth depth, camera motion, or human parameters.

Table 7: v0.3 training-configuration placeholders. Every red entry must be replaced with a verified server configuration.

Item	Configuration
Training/validation split	To be evaluated
Input resolution and clip length	To be evaluated
Optimizer / learning rate	To be evaluated
Scheduler / warm-up	To be evaluated
Batch size / accumulation	To be evaluated
Epochs or steps	To be evaluated
Mixed precision / gradient clipping	To be evaluated
GPU, training time, and random seed	To be evaluated
Checkpoint selection	To be evaluated

D.2 COMPLETE LOSSES AND WEIGHTS

The scale-stage objective supervises the residual log-scale, depth bias, and calibrated depth:

$$\mathcal{L}_{\text{scale}} = \lambda_s \mathcal{L}_{\text{log-scale}} + \lambda_b \mathcal{L}_{\text{bias}} + \lambda_d \mathcal{L}_{\text{depth}}. \quad (13)$$

Training uses metric BEDLAM depth and SMPL observations. After an initial residual-scale training stage, the accepted scale-head training path perturbs metric depth by an absolute scale factor, estimates an analytic initialization from the perturbed depth, and injects an additional initialization

error. The residual target is the ratio between the required absolute correction and the initialization actually used. Identity and non-identity perturbation groups expose the heads to both correct and inaccurate initial scales. Frames without valid scale supervision are excluded from the scale and depth losses. The bias target is zero in this training path; a general learned correction for arbitrary additive depth errors is not established by that supervision alone.

TRSTR training includes clean, scene-scale perturbation, human-translation perturbation, and mixed states. Translation perturbations act along the camera ray and tangent directions. The training objective is

$$\begin{aligned} \mathcal{L}_{\text{TRSTR}} = & \lambda_{\text{vote}} \mathcal{L}_{\text{vote}} + \lambda_{\text{trans}} \mathcal{L}_{\text{trans}} + \lambda_{\text{gate}} \mathcal{L}_{\text{gate}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} \\ & + \lambda_{\text{strong}} \mathcal{L}_{\text{strong}} + \lambda_{\text{id}} \mathcal{L}_{\text{identity}} + \lambda_{\text{safe}} \mathcal{L}_{\text{no-worse}} + \lambda_{\text{mono}} \mathcal{L}_{\text{monotonic}}. \end{aligned} \quad (14)$$

Regional votes are supervised toward the remaining target displacement divided by the number of remaining iterations. The final root estimate receives translation supervision, gates receive validity supervision, and update regularization controls correction magnitude. State-specific terms emphasize recovery from large errors, penalize displacement of clean inputs, and penalize worsening relative to the initialization or the preceding iteration. These are training penalties, not guarantees of improvement on every test example.

Placeholder: Loss weights must not be filled with expected values; they must be extracted from training configurations and checkpoint records.

D.3 TRAINING STAGES AND FREEZING STRATEGY

In the accepted scale-head training record, only the residual scale and bias heads are updated; the scene backbone, human observer, HSI interaction backbone, and body-parameter update branches remain frozen. During TRSTR training, the scale module and upstream observers remain frozen, while the regional features, translation votes, uncertainty estimates, and gates are trained. The geometric output being corrected is root translation, not pose or shape.

Both stages must report the training and validation sources, sampling ratios, perturbation ranges, effective configuration, and checkpoint-selection criterion. Local configuration inspection does not establish which checkpoints generated the reported benchmark results; this association remains to be checked against the server records. No placeholder in the tables above supplies missing experimental evidence.

E COMPLETE EVALUATION PROTOCOLS

For EMDB-2, we evaluate world-space human motion in 100-frame segments following Human3R (Chen et al., 2026). WA-MPJPE uses segment-wide similarity alignment, W-MPJPE uses alignment estimated from the first two frames, and RTE measures root-trajectory error after rigid alignment, normalized by the reference trajectory length.

Table 8: Summary of evaluation tasks and alignment conventions.

Dataset	Task	Metrics	Alignment/GT use	Status
3DPW	local human	PA-MPJPE, MPJPE, PVE	observer reference; pelvis / similarity	reference comparison
EMDB-2	world motion	WA-MPJPE, W-MPJPE, RTE	100-frame segments; 24 joints	25 sequences
TUM-Dynamic	camera trajectory	ATE	Sim(3); sequence mean	8 sequences
Bonn	video depth	Abs Rel, $\delta < 1.25$	aligned and unaligned settings	5 sequences

F ADDITIONAL QUALITATIVE RESULTS, FAILURE CASES, AND RESPONSIBLE USE

Figures 3, 5, and 6 show temporal reconstruction, human placement, and selected length measurements in the main paper. Camera and metric-depth diagnostics appear in Fig. 4 of Sec. 4.2. These examples do not replace controlled robustness or failure-rate evaluation.

The main failure chains include insufficient anchor support caused by invisibility or severe truncation; biased metric priors caused by incorrect SMPL shape or height; unreliable local probes caused

by VGGT- Ω relative-structure errors; failed environment/other-human separation under dynamic objects or multi-person occlusion; and the inability of translation-only TRSTR to correct pose, shape, or fine-grained contact.

These conditions motivate further evaluation of failure detection and reliability under missing or misleading geometric evidence.

Human-video reconstruction may be misused for privacy-sensitive behavioral analysis or surveillance. This work is limited to geometric reconstruction and does not perform identity recognition or sensitive-attribute inference. Canonical human dimensions may also produce systematic scale bias for children, uncommon body types, or populations outside the training distribution; quantifying this bias requires a dedicated sensitivity evaluation that is not supplied by the current results.